Principio de superposición lineal

Dos ondas propagantes $\Psi_1(x,t)$, $\Psi_2(x,t)$ son soluciones de

$$\frac{\partial^2\Psi_1(x,t)}{\partial x^2}=\frac{1}{v^2}\frac{\partial^2\Psi_1(x,t)}{\partial t^2}$$

$$\frac{\partial^2\Psi_2(x,t)}{\partial x^2}=\frac{1}{v^2}\frac{\partial^2\Psi_2(x,t)}{\partial t^2}$$

Una combinación lineal entre ellas $\Psi(x,t)=c_1\Psi_1(x,t)+c_2\Psi_2(x,t)$,

¿es solución de $\dfrac{\partial^2\Psi(x,t)}{\partial^2 x}=\dfrac{1}{v^2}\dfrac{\partial^2\Psi(x,t)}{\partial t^2}$?

$$\frac{\partial^2\Psi(x,t)}{\partial^2 x}=\frac{\partial^2\left(c_1\Psi_1(x,t)+c_2\Psi_2(x,t)\right)}{\partial x^2}\overset{(I)}{=}c_1\frac{\partial^2\Psi_1(x,t)}{\partial x^2}+c_2\frac{\partial^2\Psi_2(x,t)}{\partial x^2}\overset{(II)}{=}$$

$$c_1\left[\frac{1}{v^2}\frac{\partial^2\Psi_1(x,t)}{\partial t^2}\right]+c_2\left[\frac{1}{v^2}\frac{\partial^2\Psi_2(x,t)}{\partial t^2}\right]=\frac{1}{v^2}\left[c_1\frac{\partial^2\Psi_1(x,t)}{\partial t^2}+c_2\frac{\partial^2\Psi_2(x,t)}{\partial t^2}\right]\overset{(I)}{=}$$

$$\frac{1}{v^2}\frac{\partial^2\left[c_1\Psi_1(x,t)+c_2\Psi_2(x,t)\right]}{\partial t^2}=\frac{1}{v^2}\frac{\partial^2\Psi(x,t)}{\partial t^2}$$ ✓

(II) Usando las ecs.de onda de cada Ψ

(I) La derivada es una operación lineal